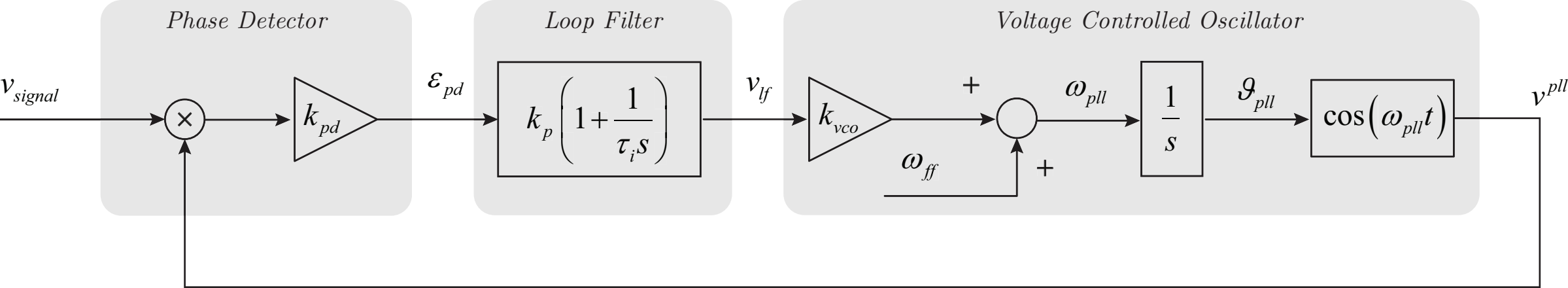
Phase Detector
Loop Filter
Voltage Controlled Oscillator
v_{signal}
k_{pd}
ε_{pd}
$k_p\left(1+\frac{1}{\tau_i s}\right)$
v_{lf}
k_{vco}
+
ω_{ff}
+
ω_{pll}
$\frac{1}{s}$
ϑ_{pll}
$\cos\left(\omega_{pll}t\right)$
v^{pll}